

CODE FOR WEBCAM TO SCAN QR CODE

```
import cv2

import numpy as np

import pyzbar.pyzbar as pyzbar

import time


# Start webcam
cap = cv2.VideoCapture(0)


# Open attendance file in append mode
attendance_file = 'attendance.txt'
names = set() # Use a set to store unique names


# Load existing names from the attendance file if it exists
try:
    with open('attendance.txt' 'r') as f:
        for line in f.readlines():
            names.add(line.strip())
except FileNotFoundError:
    print(f"File '{attendance_file}' not found. Creating new one.")


print('Reading QR codes...')


# Function to enter data into the attendance file
def enter_data(name):
    if name not in names:
        names.add(name)
        with open(attendance_file, 'a') as f:
```

```

        f.write(name + '\n')
    print(f'Added {name} is present.')
else:
    print(f'{name} is present.')

# Function to check if data is already in attendance
def check_data(data):
    name = data.decode('utf-8')
    if name not in names:
        enter_data(name)
    else:
        print(f'{name} is already in present.')

while True:
    ret, frame = cap.read()
    if not ret:
        print("Failed to capture frame.")
        break

    decoded_objects = pyzbar.decode(frame)

    for obj in decoded_objects:
        check_data(obj.data)

    cv2.imshow('Frame', frame)

    # Close window if 'q' is pressed
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

```

```
# Release the camera and close all windows
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

```
# Print final attendance list
```

```
print("\nFinal Present List:")
```

```
for name in names:
```

```
    print(name)
```

```
# Close the file after the loop
```

```
print(f"Attendance file '{attendance_file}' updated.")
```